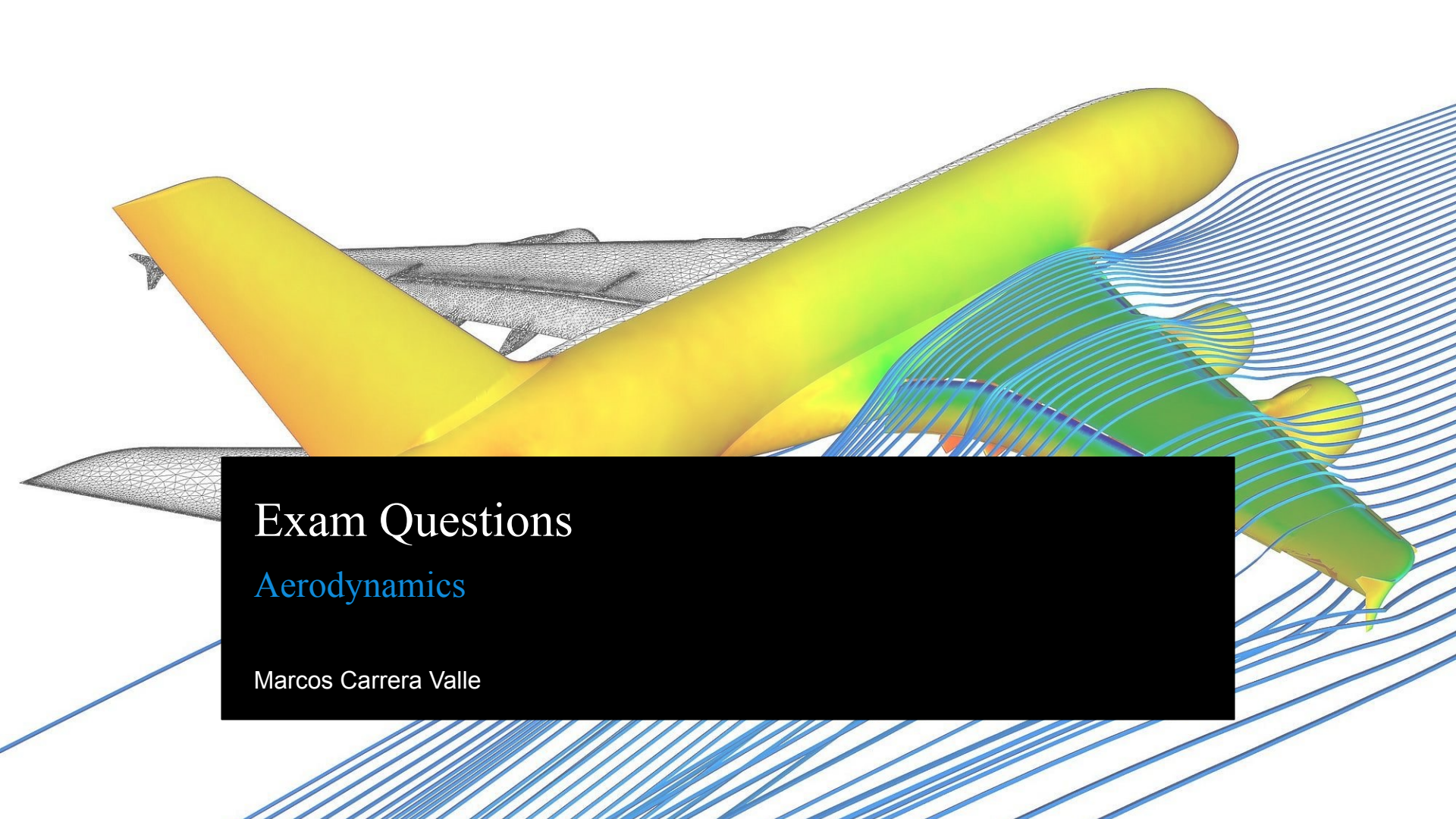


The background of the slide features a 3D visualization of an aircraft's wing and tail section. The surfaces are rendered with a color gradient representing the pressure coefficient, ranging from yellow (high pressure) to blue (low pressure). Streamlines, shown as blue lines, illustrate the flow of air around the aircraft, curving around the wing and tail. The wing is positioned diagonally across the frame, and the tail section is visible behind it.

# Exam Questions

## Aerodynamics

Marcos Carrera Valle

# Exam Questions

1. What is the SI unit of pressure?

- A) Newton
- B) Pascal
- C) Joule
- D) Watt

# Exam Questions

2. What does the equation of state relate for a perfect gas?
- A) Velocity, pressure, and temperature
  - B) Pressure, density, and temperature
  - C) Density, velocity, and pressure
  - D) Energy, temperature, and pressure

# Exam Questions

3. The Bernoulli equation is derived from which fundamental principle?
- A) Conservation of energy
  - B) Conservation of mass
  - C) Newton's second law
  - D) The equation of state

# Exam Questions

4. The continuity equation is based on which fundamental principle?
- A) Conservation of mass
  - B) Conservation of momentum
  - C) Conservation of energy
  - D) The first law of thermodynamics

# Exam Questions

5. Which of the following is NOT an assumption in the Euler equation?

- A) Steady flow
- B) No viscosity
- C) Incompressible flow
- D) No external force

# Exam Questions

6. What is specific volume?
- A) The volume of 1 kg of a substance
  - B) The inverse of velocity
  - C) The volume per unit area
  - D) The weight of a fluid per unit volume

# Exam Questions

7. Which of these is an assumption for Bernoulli's equation?

- A) Viscous effects are included
- B) The flow is unsteady
- C) The flow is incompressible
- D) External forces dominate



# Exam Questions

8. What is the stagnation point in a flow?
- A) A point where velocity is zero
  - B) A point where density is constant
  - C) A point where pressure equals atmospheric pressure
  - D) A point where mass flow rate is zero

# Exam Questions

9. The equation of state for an ideal gas is given as:

A)  $P = \rho RT$

B)  $P = RT$

C)  $P = VRT$

D)  $P = \rho VT$

# Exam Questions

10. What is the SI unit of density?

- A) kg/m
- B) kg/m<sup>2</sup>
- C) kg/m<sup>3</sup>
- D) g/cm<sup>3</sup>

# Exam Questions

11. What does the continuity equation express?

- A) Energy conservation
- B) Mass conservation
- C) Pressure variation
- D) The relationship between temperature and pressure

# Exam Questions

12. Which of the following variables is NOT involved in Bernoulli's equation?

- A) Pressure
- B) Velocity
- C) Density
- D) Temperature

# Exam Questions

13. What assumption is NOT required for the Bernoulli equation?

- A) The flow is inviscid
- B) The flow is steady
- C) The flow is compressible
- D) The flow is along a streamline

# Exam Questions

14. Which equation is used to measure an aircraft's speed using a pitot tube?
- A) The equation of state
  - B) Euler's equation
  - C) Bernoulli's equation
  - D) Continuity equation

# Exam Questions

15. If the velocity of a fluid increases, what happens to the pressure according to Bernoulli's principle?

- A) It increases
- B) It decreases
- C) It remains constant
- D) It depends on density



# Exam Questions

16. A can of coke is pressurized at 3 atm. The top surface has an area of  $0.00283 \text{ m}^2$ . What is the force exerted by the pressure inside the can?

- A) 101.5 N
- B) 85.2 N
- C) 283.7 N
- D) 860 N

# Exam Questions

17. The density of air at standard sea-level conditions is approximately:

A)  $1.225 \text{ kg/m}^3$

B)  $0.9 \text{ kg/m}^3$

C)  $1.5 \text{ kg/m}^3$

D)  $2.0 \text{ kg/m}^3$

# Exam Questions

18. The air temperature in Amsterdam is  $3^{\circ}\text{C}$  at sea level. What is the air density?

- A)  $1.23 \text{ kg/m}^3$
- B)  $1.20 \text{ kg/m}^3$
- C)  $1.25 \text{ kg/m}^3$
- D)  $1.28 \text{ kg/m}^3$

# Exam Questions

19. A stream tube has an inlet velocity of 34 m/s and a radius of 2 m. At the exit, the radius increases to 2.03 m. What is the exit velocity?

- A) 34.0 m/s
- B) 33.0 m/s
- C) 32.0 m/s
- D) 30.5 m/s

# Exam Questions

20. A Pitot tube measures a total pressure of 71000 Pa at 7000 m altitude where the static pressure is 41060.35 Pa. What is the aircraft's velocity?

(Assume  $\rho = 0.5 \text{ kg/m}^3$ )

- A) 243 m/s
- B) 280 m/s
- C) 315 m/s
- D) 346 m/s

# Exam Questions

21. Air flows through a pipe with an initial velocity of 10 m/s and a diameter of 0.5 m. If the diameter decreases to 0.25 m, what is the velocity at the narrower section?

- A) 20 m/s
- B) 30 m/s
- C) 40 m/s
- D) 10 m/s

# Exam Questions

22. Water ( $\rho = 1000 \text{ kg/m}^3$ ) flows through a horizontal pipe. If the velocity at point 1 is 5 m/s and the pressure is 200 kPa, what is the pressure at point 2 where the velocity is 10 m/s?

- A) 100 kPa
- B) 150 kPa
- C) 163 kPa
- D) 125 kPa

# Answers



# Exam Questions

1. What is the SI unit of pressure?

A) Newton

**B) Pascal**

C) Joule

D) Watt

# Exam Questions

2. What does the equation of state relate for a perfect gas?

A) Velocity, pressure, and temperature

**B) Pressure, density, and temperature**

C) Density, velocity, and pressure

D) Energy, temperature, and pressure

# Exam Questions

3. The Bernoulli equation is derived from which fundamental principle?

- A) Conservation of energy
- B) Conservation of mass
- C) Newton's second law**
- D) The equation of state

# Exam Questions

4. The continuity equation is based on which fundamental principle?

- A) Conservation of mass**
- B) Conservation of momentum
- C) Conservation of energy
- D) The first law of thermodynamics

# Exam Questions

5. Which of the following is NOT an assumption in the Euler equation?

A) Steady flow

B) No viscosity

**C) Incompressible flow**

D) No external force

# Exam Questions

6. What is specific volume?

- A) The volume of 1 kg of a substance**
- B) The inverse of velocity
- C) The volume per unit area
- D) The weight of a fluid per unit volume

# Exam Questions

7. Which of these is an assumption for Bernoulli's equation?

A) Viscous effects are included

B) The flow is unsteady

**C) The flow is incompressible**

D) External forces dominate

# Exam Questions

8. What is the stagnation point in a flow?

**A) A point where velocity is zero**

B) A point where density is constant

C) A point where pressure equals atmospheric pressure

D) A point where mass flow rate is zero



# Exam Questions

9. The equation of state for an ideal gas is given as:

**A)  $P = \rho RT$**

B)  $P = RT$

C)  $P = VRT$

D)  $P = \rho VT$

# Exam Questions

10. What is the SI unit of density?

A) kg/m

B) kg/m<sup>2</sup>

**C) kg/m<sup>3</sup>**

D) g/cm<sup>3</sup>

# Exam Questions

11. What does the continuity equation express?

A) Energy conservation

**B) Mass conservation**

C) Pressure variation

D) The relationship between temperature and pressure

# Exam Questions

12. Which of the following variables is NOT involved in Bernoulli's equation?

- A) Pressure
- B) Velocity
- C) Density
- D) Temperature**

# Exam Questions

13. What assumption is NOT required for the Bernoulli equation?

- A) The flow is inviscid
- B) The flow is steady
- C) The flow is compressible**
- D) The flow is along a streamline

# Exam Questions

14. Which equation is used to measure an aircraft's speed using a pitot tube?

- A) The equation of state
- B) Euler's equation
- C) Bernoulli's equation**
- D) Continuity equation

# Exam Questions

15. If the velocity of a fluid increases, what happens to the pressure according to Bernoulli's principle?

A) It increases

**B) It decreases**

C) It remains constant

D) It depends on density

## Exam Questions

16. A can of coke is pressurized at 3 atm. The top surface has an area of  $0.00283 \text{ m}^2$ . What is the force exerted by the pressure inside the can?

- A) 101.5 N
- B) 85.2 N
- C) 283.7 N
- D) 860 N**



# Exam Questions

17. The density of air at standard sea-level conditions is approximately:

**A) 1.225 kg/m<sup>3</sup>**

B) 0.9 kg/m<sup>3</sup>

C) 1.5 kg/m<sup>3</sup>

D) 2.0 kg/m<sup>3</sup>

# Exam Questions

18. The air temperature in Amsterdam is  $3^{\circ}\text{C}$  at sea level. What is the air density?

- A)  $1.23 \text{ kg/m}^3$
- B)  $1.20 \text{ kg/m}^3$
- C)  $1.25 \text{ kg/m}^3$
- D)  $1.28 \text{ kg/m}^3$**

# Exam Questions

19. A stream tube has an inlet velocity of 34 m/s and a radius of 2 m. At the exit, the radius increases to 2.03 m. What is the exit velocity?

- A) 34.0 m/s
- B) 33.0 m/s**
- C) 32.0 m/s
- D) 30.5 m/s

# Exam Questions

20. A Pitot tube measures a total pressure of 71000 Pa at 7000 m altitude where the static pressure is 41060.35 Pa. What is the aircraft's velocity?

(Assume  $\rho = 0.5 \text{ kg/m}^3$ )

A) 243 m/s

B) 280 m/s

C) 315 m/s

**D) 346 m/s**

## Exam Questions

21. Air flows through a pipe with an initial velocity of 10 m/s and a diameter of 0.5 m. If the diameter decreases to 0.25 m, what is the velocity at the narrower section?

- A) 20 m/s
- B) 30 m/s
- C) 40 m/s**
- D) 10 m/s

# Exam Questions

22. Water ( $\rho = 1000 \text{ kg/m}^3$ ) flows through a horizontal pipe. If the velocity at point 1 is 5 m/s and the pressure is 200 kPa, what is the pressure at point 2 where the velocity is 10 m/s?

- A) 100 kPa
- B) 150 kPa
- C) 163 kPa**
- D) 125 kPa